



H  L  NE ARVIS

PhD candidate in Operations Research

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TOOLS

Python

Java

Gurobi

CPLEX

GitHub

SQL

Latex

C++

AMPL

PROJECTS

Collective Scientific Project:

A group project to simulate and analyze optimal trajectories for racecars following the rules of the game *Racetrack*; Python

Applied Maths projects:

Study and representation of random walks, Jupyter; Study and simulation of a double pendulum, Jupyter; Study of the max-flow/min-cut problem for multi-commodity flows, Python

Industrial collaboration: Case study of optimal placement of electric vehicle charging stations in cooperation with Siemens

LANGUAGES

English: Native speaker

French: Native speaker

Spanish: Intermediate speaker

German: Beginner

INTERESTS

Sports: Competitive basketball (captain), soccer (captain)

Music: Classical piano

Associations: Vice-president of the photography association; campaign to join the school's student association; member of the Sports Office and organiser of the *Coupe de l'X* (sports tournament with 2000 participants)

Humanitarian work: Leader of a project to finance and build houses for a poor village near Chennai, India, in the summer of 2019

EXPERIENCE

PhD | EDF/INRIA

June 2024 - present

Palaiseau/Grenoble, France

- Study of aggregation in long-term models, making use of mean-field models and linear programming techniques



Master Thesis | OuiHelp

October 2023 - May 2024

Lyon, France

- Home Care Routing and Scheduling in a Mandatory Setting
- Study of the home care problem and implementation of an algorithm to improve caregiver schedules for a home care company, Python and SQL



Research internship | Mercedes-Benz AG

March 2022 - August 2022

Sindelfingen, Germany

- Research project on establishing a mixed-integer programming model for body-in-white factory planning with a modular layout, implementation in Python.
- Continuation as a working student October 2022 - March 2023



Engineering internship | SNCF R  seau - DGEX

June - August 2021

La Plaine-Saint Denis, France

- Worked on a program to optimize train schedules; analyzing the linear programming model and coding the modifications in Java



EDUCATION

Technische Universit  t M  nchen | Master's degree

Oct 2022 - May 2024

Munich, Germany

- Master's degree *Mathematics in Operations Research*



  cole Polytechnique | Top French engineering school

Sept 2019 - Sept 2022

Palaiseau, France

- Master's degree in engineering, major in Applied Mathematics
- Dipl  me de l'  cole Polytechnique
- Outstanding Investment



Lyc  e Sainte-Genevi  ve | Classe pr  paratoire

Sept 2017 - August 2019

Versailles, France

- MPSI - MP* (Math and Physics)



PUBLICATIONS

- Co-author CARV 2023: *Mixed-Integer Programming and State-of-the-Art Heuristic Approaches for the Scheduling of Modular Automotive Body-in-White Production*
DOI: 10.1007/978-3-031-34821-1_25
- Co-author SCAP 2022: *Mixed-Integer Programming Model for Scheduling of Modular Automotive Body-In-White Production Systems*
DOI: 10.1007/978-3-031-27933-1_7